

Value Chain

Decision Making Guide

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Case Overview & Management Note

The case provided in this game is of a Mobile company which manufactures and sells handsets in the Indian market. The company has decades of experience but with the recent competition, they are facing challenges to optimize cost and maximize profit. The points to be emphasized in case are:

- Market growth
- Two mobile handsets models to be launched and their consumer segments
- The sales cycle of mobile handsets
- The competitive environment
- Demand forecast

Product Feature

The participants are provided 4 product features for each of the mobile handsets i.e., premium camera, premium display, waterproof & extra memory. The participants have to decide which feature they want to apply to the handsets. While changing the product features, the following parameters change:

- Demand Estimates in the graph
- Unit Price & cost; thereby profit

With the increase in features, the demand for the product increases for the mobile duo handset, while for the mobile neo handset, the demand doesn't increase at the same pace due to its target consumer segments. The profitability also increases with increasing features but then risk increases substantially too due to higher costs and not knowing the certainty of sales. The focus should be optimizing price and cost as per mentioned consumer segments.

Demand estimates

The participants have to estimate the average monthly demand (across 8 months). The graphs provide a data point to estimate the monthly demand as per product features. With the help of a graph and demand estimate provided in the notes, a participant can decide on the forecast.

The participant is forecasting demand over 8 month cycle and hence it's necessary to provide the deviation which can be there over a period. The deviation helps the production team to formalize the plan as per the production capacity. If demand estimates go wrong then it could affect:

- Production planning
- Opportunity loss (demand > sales)

- Inventory holding cost increases
- Markdown revenue increases

Inventory holding cost

If the mobile is produced in a particular month and the team is not able to sell in that particular month due to overestimation of demand then inventory needs to be kept in the warehouse. The inventory holding cost represents the warehousing cost and opportunity cost of money as inventory gets carried over in the next month.

Markdown revenue

If a mobile handset is not sold in 8 months cycle then it has to be sold in the market at Markdown prices. The markdown prices for high-end mobile (duo) will be lesser than basic mobile (neo). The markdown prices directly indicate the loss ($\text{Cost} - \text{Markdown price} = \text{loss}$). To reduce markdown prices demand has to be realistically estimated by the use of deviation.

Opportunity loss

The term represents the money left on the table as demand is greater than sales. This is due to underestimation of demand or less production capacity.

Production Planning

In production planning, the participants have to choose between the two plants option. The difference between plants A & B is on below mentioned factors:

- Capacity (Plant B is higher)
- Outsourcing Cost (Plant A is higher)
- Transportation Cost (Plant B is higher)

The participants have to see it as an optimization issue where they have to think about demand estimates and matching production capacity. If the participants are making feature less products and demand estimation are less they can use Plant A while in another case Plant B needs to be used.

The graph in production planning represents the production choice of the company based on your demand and its deviation. The planning and capacity will change depending on the plant type.

The outsourcing will only take care of production planning based on demand estimates. For example, if 72k is plant capacity and production has planned 76k to be produced as per

demand estimates then 4k will be the outsourced capacity. If in any case due to uncertainty the actual demand is 77k then there will be an opportunity loss of 1k

Channel

The participants got two options for channels i.e., Generalist (Retail stores) & Online. The difference between the two channels are:

- Margins (Generalist is higher)
- Product Return (Online is higher)

The product return constitutes of variable which is not affected by the decision but rather by human behaviour, hence the average return would be 4% but will increase if there is a channel mismatch with consumers. Consumers preference channels for buying high-end mobiles (duo) from generalists while basic (neo) from the online channel. On basis of consumer preference and margins, capacity provided to channels can be decided.

Report Analysis

The report provides the details of the following:

- Monthly sales & demand (if demand > sales then opportunity loss)
 - This is due to forecast error or production planning error
 - It can be because of uncertainty in demand
- The outsourcing due to the difference between production planning and the available capacity of the plant chosen
- The income statement which describes the revenue & cost
 - Revenue is Product Sales + Markdown sales
 - Cost consists of following:
 - Variable production cost (mentioned on the demand planning page which increases with features)
 - Outsourcing cost (mentioned on the production planning page)
 - Inventory holding (mentioned on the demand planning page)
 - Transportation (mentioned on the production planning page)
 - Product return (4% on average and increase if mismatch in consumer segment and product)
 - Channel cost (margins mentioned on the production planning page)

Goal

To maximize the profit of Model Neo + Model Duo

General Strategies

- Focus on input and output mismatch, why it happen?
- Identify the reason was it forecast? Production?
- Optimizing the cost by considering it as a mathematical problem
- Maximization of profit by understanding the linking of features with price, cost and other parameters

Technical Aspect

Once the participant registers on the platform, the simulation game will be visible in his account.

- To play, the participant has to click on the card.
- Once all decisions are made on the particular page click on “Submit & Next/Check”, and the participant will move ahead to the next page.
- The participant can go back to the previous page again but it won't be editable
- Once all decisions are made and participants submit, the results can be viewed
- On the results page, there is an “Exit” button, which pressed will record the results and take the participant to the home page for the next iteration (if any).
- The participants can download the result into excel of any iteration at any point in time.